



DISTRIBUTED SOVEREIGN AI
INFRASTRUCTURE NODES TECHNOLOGY

FINANCIAL MODEL OVERVIEW

BUILDING GERMANY'S
SOVEREIGN AI INFRASTRUCTURE
BACKBONE

A sustainable financial
framework for long-term
value creation and
sovereign impact.



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OWNERSHIP



FINANCIAL
SUSTAINABILITY



STRATEGIC
INVESTMENT



COMMUNITY
VALUE



CAPITAL EFFICIENCY

Optimized investment
structure for maximum
return and resilience.



RISK MANAGEMENT

Robust financial modeling
with comprehensive risk
assessment.



LONG-TERM GROWTH

Scalable revenue streams
driving sustainable
profitability.



STRONG RETURNS

Attractive IRR and ROI
with solid cash flow
generation.



SOVEREIGN IMPACT

Financial success
reinforcing Europe's digital
sovereignty.



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DSAIN

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FINANCIAL MODEL OVERVIEW

Updated Strategic & Technical Edition

Version 3.0 | August 2026

Financial framework for DSAINT: validate technical performance first, validate demand and unit economics second, and scale physical infrastructure only behind evidence-based investment gates.

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Source basis: DSAINT Master Project Book v6.1, DSAINT Concept Development Document v2.0, updated DSAINT Concept/Business Model documents, Demonstrator Node 01 Masterplan, and related Bedburg-Hau infrastructure planning documents.

1. Executive Financial Summary

DSAINT is planned as an Infrastructure-as-a-Platform (IaaS) business combining physical infrastructure ownership/operation with infrastructure leasing, AI hosting, network services, energy/thermal partnerships and, over time, AI workload orchestration and trusted federation services.

The currently available DSAINT source materials contain indicative capital and operating figures, but they do not yet provide enough validated information to support a final investment-grade financial model. This document therefore separates confirmed/source-derived figures from assumptions that still require engineering, market and commercial validation.

Current financial position: sufficient for a structured feasibility model; not yet sufficient for an investment-grade financial forecast.

2. Source-Derived Financial Baseline

Source	Financial information	Status
DSAINT Master Project Book v6.1	CAPEX approximately €43M; annual OPEX approximately €6.5M	Indicative DSAINT planning baseline
DSAINT Concept Development Document v2.0	Node 01 estimated CAPEX €43M	Indicative concept estimate
AnS Datacenter Foundation / related Bedburg-Hau plans	Approximate project budget ~€40M	Alternative planning estimate with category breakdown
Updated DSAINT Business Model	Detailed CAPEX/OPEX/pricing/margin assumptions intentionally not fixed	Current strategic position

The source set contains a material difference between the €43M DSAINT estimate and the ~€40M AnS/Bedburg-Hau planning estimate. This document does not silently reconcile the figures. The current working planning range should therefore be treated as approximately €40–43M until a formal quantity-surveyed and engineered cost plan is completed.

Working planning range: ~€40–43M. Final CAPEX requires engineering, procurement and site validation.

3. Financial Model Scope

Model layer	What it should capture
Development	Feasibility, land/site, engineering, permitting and pre-construction costs
Construction	Building, electrical, cooling, security, connectivity and supporting infrastructure
Compute	GPU/server infrastructure only where DSAINT owns or finances it
Operations	Personnel, energy, connectivity, maintenance, security, insurance and administration
Platform	Orchestration, federation, monitoring and software development/operations
Revenue	Infrastructure leasing, AI hosting, managed services, orchestration/federation and other validated services
Financing	Equity, debt/project finance, strategic partners and other appropriate funding sources
Expansion	Incremental capital and operating effects of 4 MW and 6 MW stages

4. Demonstrator Node 01 CAPEX

The DSAINT Concept and Master Project Book use approximately €43M as the Node 01 development estimate. Related Bedburg-Hau infrastructure documents provide a ~€40M budget breakdown. The latter is useful as a planning structure, but it is not treated as a final DSAINT budget.

CAPEX category	Source planning amount	Comment
Land & site preparation	€5M	From AnS/Bedburg-Hau planning documents
Data center core building	€13M	From AnS/Bedburg-Hau planning documents
Cooling systems	€8M	From AnS/Bedburg-Hau planning documents
Electrical infrastructure	€9M	From AnS/Bedburg-Hau planning documents
Admin & support buildings	€3.5M	From AnS/Bedburg-Hau planning documents
Security infrastructure	€2M	From AnS/Bedburg-Hau planning documents
Engineering & permitting	€4M	From AnS/Bedburg-Hau planning documents
Optimization savings	-€4.5M	Planning adjustment in AnS documents
AnS planning total	~€40M	Source-derived planning target
DSAINt planning estimate	~€43M	Source-derived DSAINT estimate

Important: the categories above originate from related Bedburg-Hau/AnS planning documents and should not be interpreted as a validated bill of quantities for DSAINT Node 01.

5. CAPEX Items Requiring Explicit Treatment

A future investment-grade model should explicitly identify whether each item is included in the €40–43M planning range.

- GPU/server hardware
- AI accelerator financing
- High-performance storage
- Network equipment beyond facility connectivity
- Software platform development
- Initial operating working capital
- Financing fees and interest during construction
- Insurance and commissioning reserves
- Contingency
- Taxes and recoverable/non-recoverable VAT treatment
- Grid connection costs and utility deposits
- Future expansion infrastructure
- Heat-recovery infrastructure beyond the initial design

The source documents do not establish a definitive inclusion/exclusion schedule for all of these items.

6. Operating Cost Model

DSAINTE Master Project Book v6.1 identifies annual OPEX of approximately €6.5M. The source does not provide a sufficiently detailed breakdown to support a final annual operating budget.

OPEX category	Model treatment
Energy	Major variable cost; must be tied to IT load, PUE, utilization and electricity contract.
Personnel	Engineering, infrastructure operations, NOC/SOC, security, administration and management.
Connectivity	Carrier and network services.
Maintenance	Electrical, cooling, building, security, network and compute maintenance.
Software/platform	Orchestration, federation, monitoring and security platform operations.
Insurance	Infrastructure, business interruption, cyber and other relevant coverage.
Compliance	Audits, certifications, legal and regulatory activities.
Facilities	Cleaning, waste, site services and general facility operations.
Customer operations	Support, onboarding, service management and reporting.
Working capital	Cash reserve required to operate through utilization ramp-up.

Annual OPEX planning baseline: approximately €6.5M according to DSAINTE Master Project Book v6.1. This figure must be rebuilt from bottom-up operating assumptions before investment decisions.

7. Revenue Model

The DSAINT source materials identify infrastructure leasing, AI hosting, energy partnerships and network services as revenue streams. The updated business model additionally identifies managed infrastructure, orchestration and future federated compute services.

Revenue stream	Model basis	Current maturity
Infrastructure leasing	Recurring capacity/facility contracts	Concept
AI hosting	Hosting and operating customer AI infrastructure/workloads	Concept
High-density rack hosting	Secure AI-ready rack/facility services	Concept
Managed infrastructure	Operations and support services	Future
AI workload orchestration	Platform/service fee	Prototype / validation
Federated compute	Usage/capacity-based distributed compute	Future
Network services	Connectivity-related services	Concept
Energy/thermal partnerships	Potential value from energy/heat cooperation	Site dependent
Expansion leasing	Additional capacity as Node 01 expands	Future

No source document currently provides validated prices, customer contract values, utilization guarantees or take-up rates. Therefore this model does not invent revenue numbers.

8. Unit Economics Framework

The most important next financial task is to define the economics of one unit of useful AI capacity.

Metric	Formula / logic	Required input
Available IT capacity	Installed MW × operating hours	Capacity design
Useful compute capacity	Available capacity × usable utilization	Utilization assumptions
Revenue per compute unit	Customer revenue ÷ delivered compute	Validated market pricing
Energy cost per compute unit	Energy consumed × electricity price ÷ useful compute	PUE + power price
Infrastructure cost per unit	Annualized infrastructure cost ÷ useful compute	CAPEX + financing/life
Operations cost per unit	Relevant OPEX ÷ useful compute	OPEX model
Gross contribution	Revenue – variable operating costs	Validated revenue/cost data
Utilization break-even	Required utilization for operating break-even	Revenue and cost assumptions
Investment return	Cash-flow-based return metric	CAPEX + financing + cash flow

The financial model should be driven by utilization and customer demand, not by installed MW alone.

9. Utilization and Ramp-Up Model

A future financial model should use explicit utilization scenarios rather than assuming immediate full occupancy.

Scenario	Purpose	Illustrative treatment
Low utilization	Stress test	Slow customer acquisition / limited workloads
Base utilization	Planning case	Gradual ramp supported by anchor customers
High utilization	Upside case	Strong demand and effective capacity utilization
Expansion case	Growth case	2 MW → 4 MW → 6 MW subject to demand and technical gates

The percentages, prices and timing should be populated only after market validation. The current source materials do not provide defensible values for them.

10. Financing Structure

The DSAINT ownership model is based on infrastructure ownership/control rather than mandatory ownership of all compute hardware. This creates several possible financing layers.

Layer	Potential structure
Site / land	Equity, acquisition, lease or project-specific structure
Facility	Infrastructure/project finance
Power & cooling	Infrastructure financing or strategic partner arrangements
Compute hardware	DSAINTE capital, customer capital or strategic compute partners
Software platform	Founder/team capital, strategic investment, research/innovation funding where appropriate
Expansion	Reinvestment, project finance or additional strategic capital

The exact debt/equity ratio, interest rate, tenor, security package and investor return requirements are not specified in the source materials and must be developed separately.

11. Financial Development Gates

Gate	Financial question	Evidence required
G0 – Concept	Is the problem and infrastructure model credible?	Concept and ecosystem evidence
G1 – Technical validation	Can the orchestration/federation technology work?	Prototype results
G2 – Market validation	Will customers pay for the service?	Use cases, LOIs, pilots or contracts
G3 – Site feasibility	Can Node 01 be built at acceptable cost?	Engineering/site studies
G4 – Investment case	Can the project generate sustainable cash flow?	Integrated financial model
G5 – Financing	Can capital be raised on acceptable terms?	Financing offers/commitments

G6 – Construction	Is delivery controlled within budget?	EPC/procurement contracts
G7 – Operations	Does the node meet economic and operational targets?	Actual utilization, revenue and OPEX

12. Financial Strategy for the Current DSAINT Stage

The updated DSAINT strategy separates the relatively low-capital technology-validation phase from the capital-intensive physical infrastructure phase.

Stage	Financial objective
Federation Prototype 01	Minimize capital exposure while generating technical evidence.
Team integration	Acquire practical engineering capability through collaboration and employment/partnership.
Commercial validation	Test customer willingness to pay before committing major infrastructure capital.
Node 01 feasibility	Establish validated CAPEX, OPEX, demand and financing assumptions.
Node 01 construction	Deploy capital only after investment and delivery gates.
Network expansion	Replicate only after Node 01 economics are demonstrated.

This sequence reduces the risk of committing tens of millions of euros before the technology, team and market have been validated.

13. Ownership and Asset Strategy

The DSAINT business model allows infrastructure assets and compute assets to have different owners.

Asset	Preferred strategic principle
Land	DSAINTE or dedicated project ownership/control
Buildings	DSAINTE/project entity ownership or long-term control
Electrical systems	Infrastructure ownership/control
Cooling systems	Infrastructure ownership/control
Physical security	Operational responsibility
Connectivity	Contracted carrier services
GPUs/accelerators	May be DSAINT-, customer- or partner-owned
Orchestration platform	DSAINTE-controlled technology

This structure is intended to reduce exposure to rapid GPU depreciation while preserving long-term infrastructure value.

14. Financial Risks

Risk	Financial impact	Mitigation
CAPEX escalation	Higher financing requirement	Engineering, procurement validation and contingency
Energy price volatility	Higher OPEX	Long-term contracts, efficiency and energy strategy
Low utilization	Lower revenue / delayed break-even	Anchor customers and staged capacity deployment
GPU depreciation	Technology asset impairment	Partner/customer-owned compute where appropriate
Financing cost	Higher capital burden	Staged financing and project structure
Construction delay	Interest and delayed revenue	Experienced EPC and stage-gated delivery
Grid delay	Delayed commissioning	Early utility coordination
Technology change	Additional upgrade CAPEX	Modular architecture
Security/compliance events	Unexpected cost / service interruption	Security-by-design and insurance
Federation complexity	Additional platform and support costs	Controlled trusted-node rollout

15. Financial KPIs

KPI	Why it matters
CAPEX per MW	Measures infrastructure capital efficiency
Annual OPEX per MW	Measures operating efficiency
Revenue per MW	Measures monetization of infrastructure
Utilization	Primary driver of revenue efficiency
Energy cost per compute unit	Measures power burden
Gross contribution margin	Measures service economics
EBITDA / operating contribution	Measures operational viability
Cash flow	Determines funding sustainability
Break-even utilization	Shows minimum demand required
Payback / return metrics	Supports investment decisions
Debt service coverage	Relevant if project finance is used
Expansion return	Tests whether 4 MW/6 MW expansion creates value

16. Scenario Model Structure

The eventual spreadsheet model should contain at least the following linked scenarios:

1. Development case: feasibility and construction cash requirements.
2. Base operating case: expected customer ramp and recurring services.
3. Downside case: lower utilization, higher energy costs and CAPEX escalation.
4. Upside case: faster demand growth and higher utilization.
5. Expansion case: 2 MW → 4 MW → 6 MW.
6. Federation case: incremental external compute capacity with associated platform costs and revenue.

7. Financing case: alternative equity/debt/strategic-partner structures.

Each scenario should produce annual revenue, OPEX, EBITDA/operating contribution, CAPEX, financing cash flow, free cash flow, cumulative cash requirement and relevant return metrics.

17. Node 01 Expansion Economics

The Node 01 concept is designed for expansion from 2 MW toward 6 MW. Expansion should be treated as a separate investment decision rather than automatically included in the initial economic case.

Stage	Financial principle
2 MW initial	Validate customer demand, utilization, OPEX and service economics.
4 MW expansion	Proceed when incremental demand and operating economics justify additional capital.
6 MW expansion	Proceed when utilization, contracted demand, power availability and returns support expansion.

The source materials establish the capacity path but do not provide incremental CAPEX/OPEX or revenue assumptions for the 4 MW and 6 MW stages.

18. Financial Role of Strategic Partners

- Compute partners may provide GPUs/accelerators and reduce DSAINT's hardware capital exposure.
- Technology partners may contribute orchestration, software or integration capability.
- Energy partners may support power and thermal integration.
- Municipal/regional partners may contribute site and ecosystem value.
- Research partners may contribute technical validation and workloads.
- Anchor customers may provide contracted demand supporting financing.

The source documents explicitly distinguish infrastructure ownership from compute hardware ownership. This distinction is central to the DSAINT financial strategy.

19. Data Required for the Investment-Grade Financial Model

8. Validated land/site cost or lease terms.
9. Grid connection capacity, connection cost and timeline.
10. Detailed electrical design and equipment cost.
11. Detailed cooling design and cost.
12. Building area, rack density and construction quantities.
13. Confirmed scope of supporting buildings.
14. GPU/server ownership strategy.
15. Initial compute hardware quantity and procurement pricing if DSAINT-owned.
16. Electricity tariff and contract assumptions.
17. Staffing plan and salary/benefit assumptions.
18. Maintenance contracts.
19. Insurance and compliance costs.
20. Network/carrier costs.
21. Customer pricing and contract structure.

- 22. Utilization ramp and anchor customer demand.
- 23. Financing structure and terms.
- 24. Tax/VAT/accounting treatment.
- 25. Contingency and reserve requirements.
- 26. Expansion costs for 4 MW and 6 MW.

20. Conclusion

The updated DSAINT Financial Model Overview establishes a disciplined framework without pretending that unsupported financial assumptions are already known.

The current source material supports a Node 01 planning CAPEX range of approximately €40–43M and an annual OPEX baseline of approximately €6.5M from the DSAINT Master Project Book v6.1. However, the documents do not yet support validated revenue, utilization, pricing, financing or return assumptions.

The next financial milestone should therefore be an integrated bottom-up model driven by engineering data, customer demand and prototype evidence. The DSAINT financial strategy should remain stage-gated: minimize early capital exposure, validate technology and market demand, then finance physical infrastructure on the basis of evidence.

Financial principle: validate first. Finance second. Scale third.

21. Document Status

Item	Information
Project	DSAIN T – Distributed Sovereign AI Infrastructure Nodes Technology
Financial focus	Demonstrator Node 01 and future distributed infrastructure
Founder	Ernest Repko
Location	Germany / Node 01: Bedburg-Hau
Email	info@dsaint.eu
Website	www.dsaint.eu
Document	Financial Model Overview – Updated Strategic & Technical Edition
Version	3.0
Date	August 2026
Status	Financial framework / planning model; not an investment-grade forecast

Financial Model Overview – conceptual planning document; not an offer of securities or investment solicitation.